

Preflight hang FIX + audit pass (v0.72.3)

Why startup froze, what it does now, and what the top-to-bottom audit + mutation testing turned up.

The hang, in one line

Startup verified EVERY page of the 3.65 GB index (a full PRAGMA quick_check). Instant when the database was small; a multi-minute freeze once OCR filled it. The program wasn't broken — it was grinding through a needless full scan before the server could start.

FIX

Startup now does a millisecond open-and-read probe. Databases over 512 MB auto-skip the full scan. The deep scan is opt-in (--deep) and time-budgeted (20 s) so it can NEVER hang. The watchdog and /healthz share this code, so they got faster too.

Audit findings — 5 fixed, 3 flagged, 0 critical open

✓ Fixed

- V-01 preflight hang (full scan on 3.65 GB).
- V-02 /api/threed_refs 500 (orphaned helper) -> test_routes 34/34.
- V-03 procedure parser infinite loop (i=1) -> restored + new test.
- V-04/05 two DB connection leaks (procedure_full, threed_refs).

Additive, rollbackable.

⚠ Flagged for your call

- V-06 safeguard.py still runs a full quick_check (same hang class).
- V-07 procedure parser weakly tested (20% mutation kill).
- V-08 dev-only: sandbox mount tears big-file reads.

No code changed; awaiting sign-off.

⚡ Mutation testing

- New mutate.py: auto-breaks one token at a time, re-runs tests.
- patterns.py: 13/13 killed = 100% (tests are airtight).
- procedure_feature.py: 20% (the V-07 gap).
- Full pass on your PC via RUN-MUTATION.bat.

Survivors = test blind spots.

Verify on your PC: start the app (no more hang), run RUN-ALL-TESTS.bat (test_routes 34/34, new test_procedure), and RUN-MUTATION.bat for the full mutation pass. Report: docs/audit/AUDIT-REPORT.html + findings.json.